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Core Body Temperature and Circadian Rhythm of Hot Flashes in Menopausal Women*

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ABSTRACT

Postmenopausal hot flashes are characterized by sweating and peripheral vasodilation and occur more frequently during increased heat loads. The circadian rhythm of core body temperature (T_C) is well known and suggests that hot flashes will be most frequent when core temperature is highest. This hypothesis has not been tested previously. Ten symptomatic and six asymptomatic postmenopausal women were recruited from advertisements and screened. Each received 24-h ambulatory monitoring of sternal skin conductance levels to detect hot flashes, ambient temperature, skin temperature, and T_C. The last measure was recorded using an ingested radiotelemetry pill. Cosinor analysis demonstrated a circadian rhythm ($P < 0.02$) of hot flashes with a peak at about 1825 h. T_C values of the symptomatic women were lower than those of the asymptomatic women ($P < .05$) from 0000–0400 h and at 1500 and 2200 h. The majority of hot flashes were preceded by elevations in T_C. Thus, elevated T_C may serve as one trigger of menopausal hot flashes. (*J Clin Endocrinol Metab* **80:** 2354–2358, 1995)

HOT FLASHES (HFs) are the most common symptom of the climacteric and affect most postmenopausal women. They are experienced as sensations of internal heat in the upper body and often cause awakening during the night. HFs exhibit the physiological characteristics of a heat dissipation response, including sweating, cutaneous vasodilation, and increased heart rate (1–3). They occur more frequently in warm than in cool rooms (4) and can be provoked by peripheral heating (1).

The relationship between core body temperature (T_C) and the occurrence of HFs has not been clearly established. The circadian rhythm of T_C is well known, and similar variations in other thermoregulatory parameters, such as heat conductance and sweating, have also been shown (5, 6). These patterns suggest that the thermoregulatory effector responses of HFs might also demonstrate temporal variations. A previous study showed circadian rhythmicity of self-reported HFs in some menopausal women, but no physiological data were collected (7).

We demonstrated previously that a criterion increase in skin conductance recorded from the sternum correlates highly with self-reported HFs in ambulatory and laboratory conditions (1). We therefore recorded sternal skin conductance, ambient temperature, skin temperature (T_{SK}), and T_C using ambulatory monitoring for 24-h periods in symptomatic and asymptomatic women. This procedure enabled us to collect data on many HFs in different women. Because our patients were reluctant to wear rectal thermistor probes outside of the laboratory, we recorded T_C using a commercially available radiotelemetry pill.

Materials and Methods

Ten postmenopausal women reporting frequent HFs and six postmenopausal women reporting no HFs were recruited from newspaper advertisements requesting participants for research on menopause. All were in good health, as determined by medical history and a physical examination by a gynecologist. All were amenorrheic for at least 1 year and were free of all medication. None had received hormone replacement therapy. There were no significant differences between the two groups on any demographic characteristic (Table 1). Blood was drawn at intake for measurements of plasma LH, FSH, estradiol, and progesterone by RIA. There were no significant differences between the groups on any measure (Table 2). All women had FSH levels greater than 30 mIU/mL and estradiol levels less than 25 pg/mL. All women gave written informed consent according to procedures approved by our institutional review board.

The subjects' sternal skin conductance was recorded for a single 24-h period starting at about 1000 h. The women wore loose-fitting, cotton clothing. Skin conductance level was measured with two silver/silver chloride electrodes (Meditrace 8201, Graphic Controls, Buffalo, NY), 1.5 cm in diameter, which were filled with 0.05 mol/L KCl paste and attached over the sternum. They were connected to a 0.5-volt constant-voltage circuit and one direct current (dc) channel of an Oxford Medilog 9000 recorder (Oxford, UK). Ambient temperature was recorded on a second dc channel of the Medilog, using a Yellow Springs 731 thermistor (Yellow Springs, OH) mounted on an insulated Plexiglas pedestal attached to the wrist. T_{SK} was recorded on a third dc channel with another Yellow Springs 731 thermistor taped to the dorsal surface of the proximal phalanx of the third finger. The Medilog tapes were replayed at 20 times real time with an Oxford Medilog playback unit, an analog-to-digital converter, and a computer (IBM PC). The computer was programmed to define a hot flash as a skin conductance increase greater than or equal to 2 μmho in 30 sec. We showed previously that this sternal skin conductance response has a high correspondence with self-reported HFs and does not occur in premenopausal women during laboratory or ambulatory monitoring (1). Using these methods, hot flashes were detected in all of the symptomatic women (range 1 to 22 per 24 h). No hot flashes were detected in the asymptomatic women.

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TABLE 1. Demographic characteristics of subjects (mean ± SE)

	Symptomatic women	Asymptomatic women
Age	54.0 ± 1.0	52.3 ± 1.6
Race	W = 7	W = 5
	B = 3	B = 1
Height (cm)	162.1 ± 2.3	165.6 ± 1.5
Weight (kg)	74.7 ± 4.8	70.5 ± 4.8
Skinfold thickness (mm)		
Triceps	28.1 ± 2.6	21.8 ± 2.6
Subscapular	21.8 ± 3.8	25.1 ± 2.0
Reported bedtime (h)	2250 ± :43	2243 ± 1:14
Reported awakening time (h)	0735 ± :26	0826 ± :46

All group comparisons except race were performed using individual *t* tests. Race was compared with Fisher's exact test. There were no significant group differences on any measure.

Detailed hourly diaries of all activities were completed by the subjects. Typical activities included housework, running errands, and watching television. Subjects were instructed to refrain from physical exertion such as running, dancing, or treadmill exercise. They were instructed to note all intake of food and beverages and exposure to thermal stimuli, such as cooking on a stove.

T_C was recorded every 30 sec using an ingested radiotelemetry pill (CorTemp; Human Technologies, St. Petersburg, FL). The data were stored in memory by the CorTemp recorder, worn on a belt with the Medilog, and subsequently loaded into an IBM personal computer for further analysis.

Before usage, each pill was checked for accuracy against a thermometer certified by the National Bureau of Standards, accurate to 0.05 C, in a circulating water bath at temperatures of 35.5, 36.0, 36.5, 37.0, and 37.5 C. If the pill deviated from the standard by more than 0.1 C at any step, it was recalibrated. Additionally, we compared CorTemp and rectal temperature values in three symptomatic women in our laboratory. Rectal temperature was recorded using disposable Yellow Springs probes and one dc channel of a Grass (Quincy, MA) Model 7D polygraph. This system was calibrated using the same water bath and thermometer described above. Subjects wore cotton scrub suits and slept in a temperature- and humidity-controlled laboratory (23 C, 50% relative humidity) from 2230 to 0730 h. The average temperatures for the CorTemp and rectal measurements were 36.67 ± 0.17 C (SD) and 36.61 ± 0.18 C, respectively. The average difference between the two measurements was 0.07 C.

T_C values were graphically displayed for individual recordings and manually edited to eliminate artifacts. Approximately 90 min of data at the beginning of each recording were eliminated to allow egress of the telemetry pill from the stomach. The T_C data were averaged in 1-h blocks and analyzed using the cosinor method. Each data series was analyzed by a single cosinor involving the least-squares fit of a 24-h cosine curve to the data (8). The following rhythm parameters were obtained: the MESOR, a rhythm-adjusted mean; the circadian amplitude, a measure of the extent of predictable change within a day; and the circadian acrophase, a measure of timing of overall high values recurring each day. The residual sum of squares was used for hypothesis testing, *i.e.* for testing the zero-amplitude (no-rhythm) assumption and to obtain 95% confidence intervals for each parameter. The circadian characteristics for each woman in each group were further summarized by population-mean cosinor, a procedure examining the extent of clustering of circadian amplitude-acrophase pairs within a group (9). Rejection of the zero-amplitude (no-rhythm) assumption with this procedure implies generalization of results to the population. Parameter tests (10) were used to compare the circadian patterns of T_C between women with and without hot flashes and also to compare the timing of the circadian rhythm of T_C with that of the incidence of hot flashes for symptomatic women. Group differences in T_C and T_{SK} at specific hourly time points were tested with two-way repeated measures analyses of variance (ANOVAs) and simple effects tests.

Complete physiological data for the 30 min before and the 30 min after HF onset were obtained for 77 HFs. The T_C, ambient temperature, and skin conductance level data were analyzed using separate one-way, repeated-measures ANOVAs with Duncan post-hoc tests. The minimum level of statistical significance for all analyses was 0.05.

TABLE 2. Hormone levels of subjects (mean ± SE)

	Symptomatic Women	Asymptomatic Women
Estradiol (pmol/L)	37 ± 2	56 ± 15
Progesterone (nmol/L)	0.19 ± 0.1	0.35 ± 0.13
FSH (IU/L)	61 ± 6	76 ± 12
LH (IU/L)	21 ± 3	32 ± 14

All group comparisons were performed using individual *t* tests. There were no significant group differences on any measure.

Results

HFs were detected in all 10 symptomatic women during ambulatory monitoring and ranged in incidence from 1 to 22 per day. The distribution of HFs across the 24-h period is shown in Fig. 1. A circadian rhythm in HF incidence was demonstrated with statistical significance in 6 of the 10 women ($P < 0.05$) and with borderline significance ($P = 0.057$) in 1 additional woman (Table 3). On a population basis, the overall statistical significance of the circadian pattern of HFs is $P = 0.02$.

Figure 1 also shows the average 24-h T_C data for both groups of women and the best-fit cosine curve for each group. The T_C data were first examined with least-square spectral analysis, which indicated that the circadian rhythm was invariably the most prominent component. This rhythm was statistically significant ($P < 0.01$) for each of the 16 women; on a population basis, it was statistically significant for the symptomatic group at P less than 0.001 and for the asymptomatic group at P less than 0.008. A comparison of the circadian characteristics of the two groups showed that the MESOR (24-h mean) of T_C was significantly ($P < 0.05$) lower for the symptomatic group (36.67 ± 0.05 C) than for the asymptomatic group (36.93 ± 0.06 C). This comparison is seen most clearly in Fig. 1. There were no significant group differences in circadian amplitude or acrophase. Analysis of variance showed that group differences in T_C were significant ($P < 0.05$) at 0000–0400 h, 1500 h, and 2200 h. There were no significant effects from the ANOVA on T_{SK} (symptomatic mean = 29.0 ± 4.4 C *vs.* asymptomatic mean = 30.8 ± 3.1 C, $P > 0.4$).

To test the phase relationship between the circadian rhythm of HFs and that of T_C, it was necessary to normalize the data because HF frequency and body temperature are expressed in different units. Therefore, all MESORs were equated to 1, and the amplitudes were rescaled to achieve an acrophase-unweighted average of 1 for each group. Parameter tests showed that the difference in circadian acrophase for HFs and T_C was of borderline statistical significance ($P = 0.056$); *i.e.* the circadian phase with the largest overall number of HFs (−276° or 1825 h) tended to lag behind the circadian phase of peak core temperature (−229° or 1529 h) by about 3 h (see Fig. 1). A linear regression of the circadian acrophase as a function of the MESOR of HF incidence showed that a greater incidence of HFs was associated with a later circadian acrophase ($P = 0.016$).

Complete physiological data for the 30 min before and the 30 min after the beginning of each HF were available for 77 HFs. A one-way ANOVA on the T_C data showed a significant

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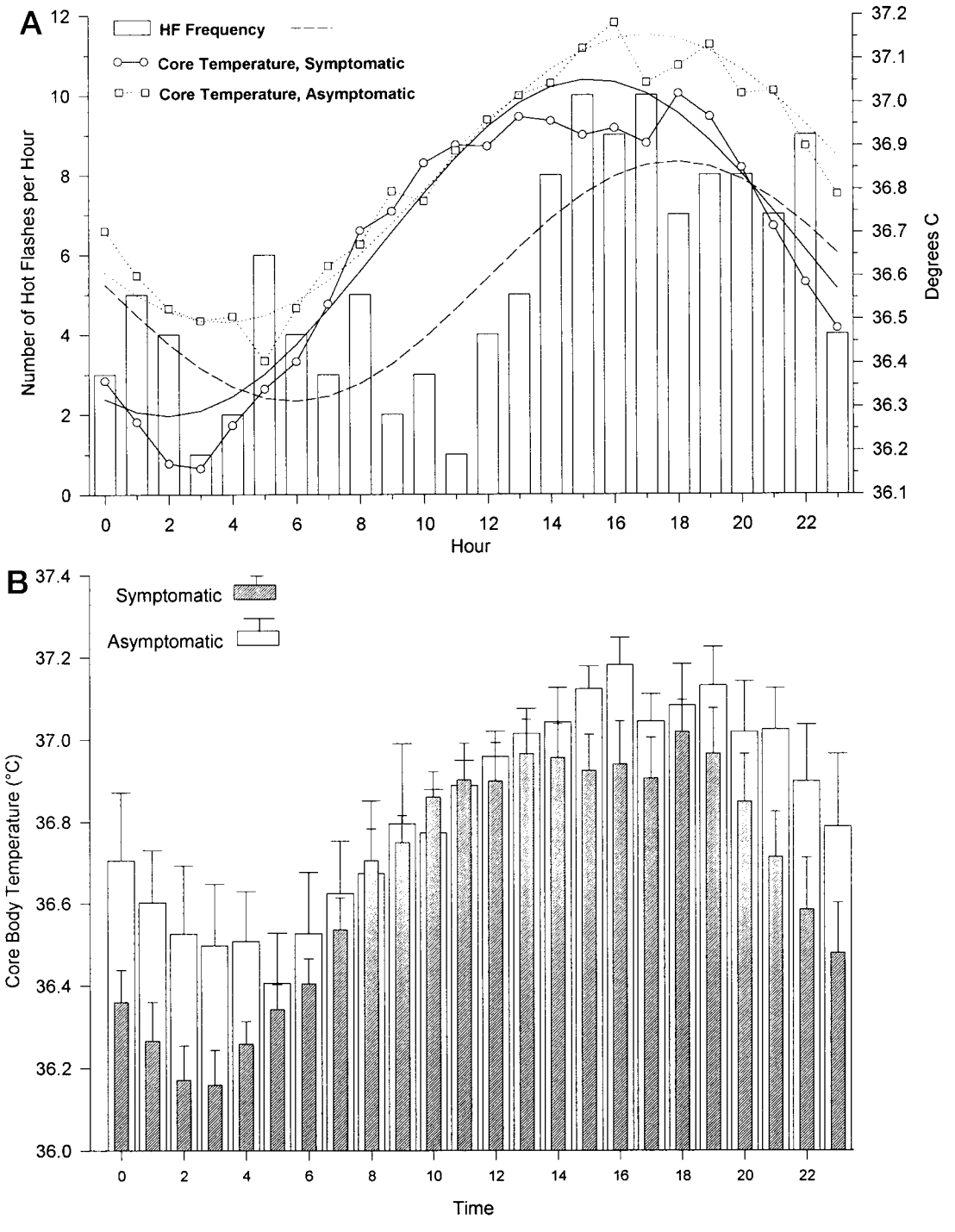


FIG. 1. A, Hot-flash frequency and T_C during 24 h. Hot-flash frequency in 10 symptomatic women (*bars*); best-fit cosine curve for hot-flash frequency (*dashed line*); 24-h T_C data for 10 symptomatic women (○) with best-fit cosine curve (*solid line*); 24-h T_C data in 6 asymptomatic women (□) with best-fit cosine curve (*dotted line*). B, 24-h T_C (mean ± SE) in symptomatic and asymptomatic women.

($P < 0.001$) epoch effect (Fig. 2A). Duncan post-hoc tests showed that the temperature increase between minutes −17 and −2 (the temperature peak) was statistically significant ($P < 0.05$), as was the temperature decline between minutes −2 and +7. Inspection of the raw temperature data between minutes −17 and −2 revealed increases in 46 HFs, declines in 23 HFs, and no change in 8 HFs. A *t* test comparing core temperature at minute −17 for the 77 HFs *vs.* all core temperatures recorded for 24 h in the symptomatic women showed that the HFs occurred at significantly ($P < 0.02$) higher temperatures (36.82 ± 0.04 C *vs.* 36.70 ± 0.005 C).

The epoch effect for skin conductance level (Fig. 2B) was significant because this measure was used to define the HFs ($P < 0.0001$). The epoch effect for ambient temperature (Fig. 2C) was also significant ($P < 0.02$) and indicated a gradual decline across the 60-min period.

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TABLE 3. Circadian rhythm characteristics of the incidence of hot flashes in menopausal women[a]

ID	PR (95% CI)	*P*	MESOR ± SE	Amplitude ± SE	Acrophase
503	50	<0.001	0.501 ± 0.109	0.705 ± 0.154	−265° (−238°, −292°)
515	37	0.008	0.417 ± 0.122	0.601 ± 0.172	−290° (−254°, −327°)
516	36	0.009	0.917 ± 0.142	0.687 ± 0.201	−322° (−284°, −359°)
518	54	<0.001	0.834 ± 0.126	0.881 ± 0.178	−321° (−297°, −346°)
519	19	0.113	0.209 ± 0.127	0.394 ± 0.179	−218° (0, 0)
520	4	0.663	0.709 ± 0.131	0.169 ± 0.185	−313° (0, 0)
521	9	0.385	0.043 ± 0.042	0.083 ± 0.059	−83° (0, 0)
522	28	0.032	0.292 ± 0.113	0.456 ± 0.160	−210° (−163°, −256°)
525	26	0.040	0.667 ± 0.103	0.401 ± 0.146	−163° (−113°, −212°)
527	24	0.057	0.459 ± 0.110	0.397 ± 0.155	−260° (0, 0)
Population summary	28.7	0.020	0.505 ± 0.100	0.305 (0.094, 0.516)	−276° (−215°, −311°)

[a] PR, percent rhythm (proportion of variability accounted for by fitted model); *P*, *P* value from test of zero-amplitude (no-rhythm) assumption; acrophase in (negative) degrees, with 360° = 24 h, 0° = local midnight; SE, standard error; 95% CI, 95% confidence interval.

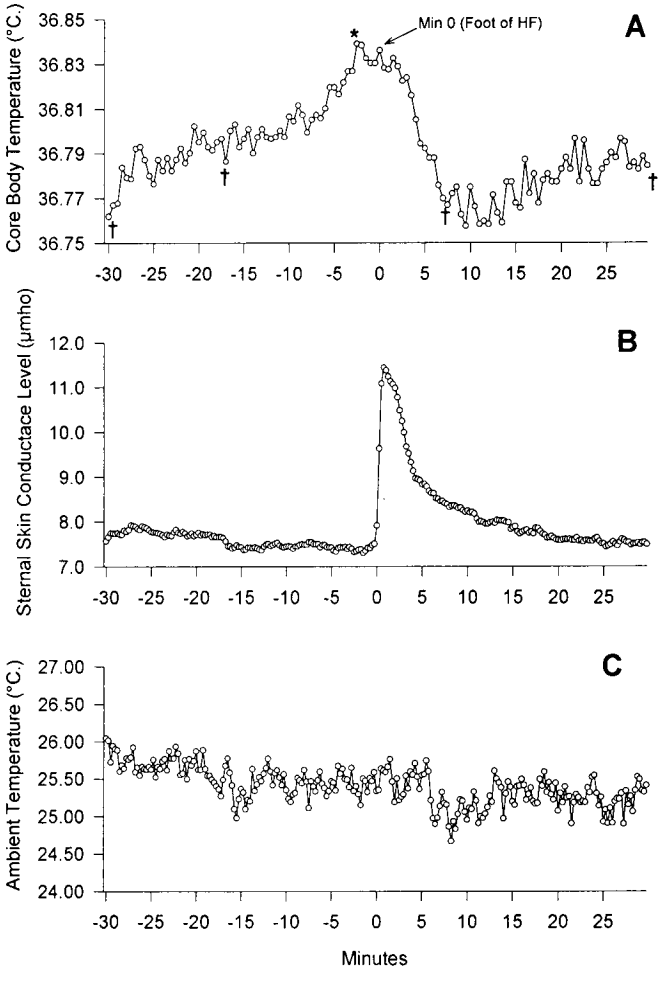


FIG. 2. A, T_C (means) surrounding 77 hot flashes. Time zero is the beginning of each hot flash determined from the skin conductance response. Points marked † are significantly different from peak temperature marked * at *P* less than 0.05. B, Sternal skin conductance levels (means) surrounding 77 hot flashes. C, Ambient temperature (means) surrounding 77 hot flashes.

Discussion

In the present study, we found a statistically significant circadian rhythm of HFs with an acrophase at 1825 h. HFs were defined objectively using a previously validated physiological measure and did not rely on self-reports by the subjects. A previous study of self-reported HFs found two temporal patterns: one with a peak frequency in the morning (0400–0959 h) and one with a peak in the evening (1600–2159 h) (7). We detected the evening but not the morning peak. Self-reports of events occurring during sleep are notoriously unreliable, and it is possible that the early morning reports of the previous study were exaggerated. Alternatively, it is possible that a morning peak of HF incidence actually exists (see our Fig. 1), but it was not significant in our relatively small sample of subjects. Finally, we considered the possibility that our skin conductance technique might not accurately record HFs. However, there is considerable research to validate this methodology. One investigation showed a correspondence of 95% between the criterion skin conductance change and subject self-reports for HFs recorded in the laboratory and 86% for those recorded during ambulatory monitoring (1). A subsequent study obtained values of 84% and 77%, respectively (11). Another study conducted in the laboratory found a correspondence of 97% between the objective and subjective methods (12). Thus, we are confident in the validity of the procedure.

We also found that T_C was significantly lower in symptomatic than in asymptomatic women at 1500 h, 2200 h, and 0000–0400 h. Previous research showed that about 1.3 g·min of sweat is lost during HFs and that evaporation of this sweat would approximately double the rate of heat loss for the duration of the HF (13). Thus, it is possible that heat lost during HFs could lower T_C at 1500 and 2200 h, which is when HFs were frequent. This reasoning is supported by work showing declines in esophageal and rectal temperatures after HFs of −0.3 and −0.7 C, respectively (3, 13). However, this explanation is unlikely to account for the T_C difference from 0000–0400 h when HFs were infrequent. T_C declines during stage 4 (slow wave) sleep (14), and we showed previously that symptomatic postmenopausal women have three times more stage 4 sleep than do asymptomatic women (15). This finding is more likely to explain the T_C difference found during sleep.

Previous investigations did not detect an increase of T_C before HFs, but different methods of detection were used, such as measurement of rectal temperature (13). Other stud-

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ies showed that rectal temperature responds slowest to changes in T_C when compared with temperatures of the stomach, intestines, colon, and pulmonary artery blood (16–18). Thus, the faster response time of our method may have detected the T_C increases preceding HFs.

The T_C elevation before HFs that was seen in the present study was small in magnitude but statistically significant. It is not an artifact of increased ambient temperature, which showed a gradual decline during the period of interest. The cause of this T_C increase is not known. We recently found a significant increase in respiratory quotient (without hyperventilation) before and during HFs, suggesting an increase in metabolic rate (19). Increased metabolic rate or decreased heat loss (or both) preceding HFs could account for the increase in T_C. Additional research on this point would be of interest.

The temporal relationship between the increases in core temperature and sternal skin conductance suggests that the former may serve as one trigger of HFs. In humans, small elevations in core temperature produce increases in peripheral temperature that are 20-fold greater (20). Additionally, it has been estimated that "a significant increase in steady-state heat loss . . . will be caused by an average increase of steady-state temperature of only 0.01 C at the central sensor" (21). The average T_C elevation preceding HFs in the present study was 0.08 C, which probably exceeds this level of sensitivity. Furthermore, peripheral temperature elevations on the order of 1.6 C (*i.e.* 20×0.08 C) are commonly seen during HFs (1–3). Thus, it is plausible that T_C elevations of the magnitude found in the present study would be sufficient to trigger HF heat-loss responses.

T_C elevations before HFs were not found in 40% of the cases and therefore cannot serve as the sole determinant. We showed previously that levels of plasma 3-methoxy-4-hydroxyphenylglycol, the main metabolite of brain norepinephrine, are significantly elevated before and during HFs (22). We also showed that injections of yohimbine, an α_2-adrenergic compound that elevates plasma 3-methoxy-4-hydroxyphenylglycol, provoke HFs (23). Other studies showed that plasma concentrations of 3-methoxy-4-hydroxyphenylglycol correlate with sweating activity (24) and that intrahypothalamic injection of norepinephrine elevates core temperature and facilitates heat-loss responses (25). Thus, T_C elevations and central noradrenergic activation may serve as two related factors that separately or together can trigger HFs.

In summary, the present study has demonstrated a significant circadian rhythm of menopausal HFs that is related to the circadian rhythm of T_C. T_C elevations occurred before the majority of HFs and may serve as one triggering mechanism. Heat loss during HFs may account for reduced T_C when HFs are most frequent, but it does not explain reduced T_C during sleep.

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